

Incident Report: nginx Fails to Start Because Port 80 Is Already in Use

Scenario 2 • Linux Web Server Troubleshooting Labs

Summary

nginx failed to start even though its configuration was valid. Another process, a Python web server (PID 2944), was already listening on TCP port 80, so nginx could not bind to it. The service was restored by identifying and stopping the conflicting process and restarting nginx, then verified by confirming nginx owned port 80 and that the HTTP response came from nginx.

This was a controlled lab exercise. The conflict was created deliberately by starting a Python web server on port 80 to stand in for an unexpected service.

Environment

- Debian 13 virtual machine (VirtualBox), hostname soc-lab
- nginx web server, managed by systemd
- Tools used: systemctl, journalctl, nginx -t, ss, curl, kill

Timeline (EDT)

Time	Event
17:26	Baseline: nginx owns port 80 and <code>curl -I</code> returns 200 OK from nginx
17:42:36	nginx stopped, Python web server started on port 80. First start attempt fails: <code>bind() to 0.0.0.0:80 failed (98: Address already in use)</code>
17:54:09	Second start attempt fails with the same <code>bind()</code> error
17:54:12	<code>systemctl status</code> shows failed (exit-code); <code>nginx -t</code> still passes
18:45:11	<code>ss -tlnp</code> shows python3 (PID 2944) on port 80. <code>curl -I</code> returns 200 OK but Server: SimpleHTTP/0.6 Python
Before 18:48	Stopped the Python process with <code>sudo kill 2944</code>
18:48:43	nginx started successfully. Port 80 owned by nginx, curl returns 200 OK from nginx

Symptom

`sudo systemctl start nginx` failed with: *Job for nginx.service failed because the control process exited with error code.* The service state was failed (Result: exit-code).

Investigation

1. Check service state. `systemctl status nginx --no-pager` showed Active: failed. The config test step (ExecStartPre, `nginx -t`) exited with **status=0/SUCCESS**, but the start step (ExecStart) exited with **status=1/FAILURE**. This is the key difference from a config error: the configuration passed, but nginx still could not start.

2. Read the logs. `sudo journalctl -u nginx -n 20 --no-pager` showed repeated `[emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)` followed by `still could not bind()`. Something else already held the port.

3. Confirm the config is valid. `sudo nginx -t` reported syntax is ok and test is successful, ruling out a configuration problem.

4. Find what owns the port. `sudo ss -tlnp | grep :80` showed python3 (PID 2944) listening on `0.0.0.0:80`. `curl -I http://127.0.0.1` returned 200 OK, but the `Server: header` read `SimpleHTTP/0.6 Python/3.13.5`, not nginx. The site looked healthy, but the wrong server was answering.

Root Cause

A different process (Python `http.server`, PID 2944) was already bound to TCP port 80. Only one process can listen on a given address and port, so nginx's `bind()` call failed with error 98 (address already in use) and the service exited. The nginx configuration was correct throughout.

Resolution

Stopped the conflicting process, confirmed port 80 was free, and started nginx:

```
sudo kill 2944
sudo ss -tlnp | grep :80      # no output = port free
sudo systemctl start nginx
```

Verification

- `sudo ss -tlnp | grep :80`: port 80 owned by nginx (master and worker processes)
- `systemctl status nginx --no-pager`: active (running); `ExecStartPre` and `ExecStart` both `status=0/SUCCESS`
- `curl -I http://127.0.0.1`: HTTP/1.1 200 OK with `Server: nginx`

Prevention

- Check what is listening before starting a service: `sudo ss -tlnp`.
- Document which service is supposed to own each port, and alert when an expected service is down or a port is held by an unexpected process.
- Verify the response actually comes from the intended service (the `Server: header`), not just that something returns 200 OK.
- Treat an unexpected listener as a potential security finding: confirm what it is, who started it, and whether it should be running.

Evidence

1. Baseline: nginx owns port 80 and responds with 200 OK

```
mohammed@soc-lab:~$ sudo ss -tlnp | grep :80
LISTEN 0      511          0.0.0.0:80      0.0.0.0:*      users:(("nginx",pid=2851,fd=5),("nginx",pid=2850,fd=5))
LISTEN 0      511          [::]:80        [::]:*        users:(("nginx",pid=2851,fd=6),("nginx",pid=2850,fd=6))
mohammed@soc-lab:~$ curl -I http://127.0.0.1
HTTP/1.1 200 OK
Server: nginx
Date: Mon, 05 Oct 2026 21:26:26 GMT
Content-Type: text/html
Content-Length: 615
Last-Modified: Mon, 28 Sep 2026 23:33:06 GMT
Connection: keep-alive
ETag: "6abaf932-267"
Accept-Ranges: bytes
```

2. nginx start fails

```
mohammed@soc-lab:~$ sudo systemctl start nginx
Job for nginx.service failed because the control process exited with error code.
See "systemctl status nginx.service" and "journalctl -xeu nginx.service" for details.
mohammed@soc-lab:~$
```

3. Service status: ExecStartPre (config test) succeeded, ExecStart failed

```
mohammed@soc-lab:~$ systemctl status nginx --no-pager
* nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Mon 2026-10-05 17:54:12 EDT; 1min 27s ago
     Duration: 5min 40.613s
  Invocation: a5b56802b12f46d9bca6c0eec5515a2a
     Docs: man:nginx(8)
    Process: 3008 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 3010 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=1/FAILURE)
   Mem peak: 1.0M
      CPU: 36ms
mohammed@soc-lab:~$ _
```

4. Journal: bind() to 0.0.0.0:80 failed (98: Address already in use)

```
mohammed@soc-lab:~$ sudo journalctl -u nginx -n 20 --no-pager
[sudo] password for mohammed:
Oct 05 17:42:36 soc-lab systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Oct 05 17:42:36 soc-lab nginx[2960]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Oct 05 17:42:36 soc-lab nginx[2960]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Oct 05 17:42:37 soc-lab nginx[2960]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Oct 05 17:42:37 soc-lab nginx[2960]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Oct 05 17:42:38 soc-lab nginx[2960]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Oct 05 17:42:38 soc-lab nginx[2960]: nginx: [emerg] still could not bind()
Oct 05 17:42:38 soc-lab systemd[1]: nginx.service: Control process exited, code=exited, status=1/FAILURE
Oct 05 17:42:38 soc-lab systemd[1]: nginx.service: Failed with result 'exit-code'.
Oct 05 17:42:38 soc-lab systemd[1]: Failed to start nginx.service - A high performance web server and a reverse proxy server.
Oct 05 17:54:09 soc-lab systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Oct 05 17:54:10 soc-lab nginx[3010]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Oct 05 17:54:10 soc-lab nginx[3010]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Oct 05 17:54:11 soc-lab nginx[3010]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Oct 05 17:54:11 soc-lab nginx[3010]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Oct 05 17:54:12 soc-lab nginx[3010]: nginx: [emerg] still could not bind()
Oct 05 17:54:12 soc-lab systemd[1]: nginx.service: Control process exited, code=exited, status=1/FAILURE
Oct 05 17:54:12 soc-lab systemd[1]: nginx.service: Failed with result 'exit-code'.
Oct 05 17:54:12 soc-lab systemd[1]: Failed to start nginx.service - A high performance web server and a reverse proxy server.
mohammed@soc-lab:~$
```

5. nginx -t passes: the config is not the problem

```
mohammed@soc-lab:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
mohammed@soc-lab:~$
```

6. Culprit found: python3 (PID 2944) owns port 80, Server header is SimpleHTTP

```
mohammed@soc-lab:~$ sudo ss -tlnp | grep :80
LISTEN 0      5            0.0.0.0:80      0.0.0.0:*      users:(("python3",pid=2944,fd=3))
mohammed@soc-lab:~$ curl -I http://127.0.0.1
127.0.0.1 - - [05/Oct/2026 18:45:11] "HEAD / HTTP/1.1" 200 -
HTTP/1.0 200 OK
Server: SimpleHTTP/0.6 Python/3.13.5
Date: Mon, 05 Oct 2026 22:45:11 GMT
Content-type: text/html; charset=utf-8
Content-Length: 636
mohammed@soc-lab:~$ _
```

7. Recovery: nginx owns port 80, active (running), 200 OK from nginx

```
mohammed@soc-lab:~$ sudo ss -tlnp | grep :80
LISTEN 0      511          0.0.0.0:80      0.0.0.0:*      users:(("nginx",pid=3192,fd=5),("nginx",pid=3191,fd=5))
LISTEN 0      511          [::]:80        [::]:*         users:(("nginx",pid=3192,fd=6),("nginx",pid=3191,fd=6))
mohammed@soc-lab:~$ systemctl status nginx --no-pager
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-10-05 18:48:43 EDT; 1min 15s ago
     Invocation: 166dce8155dc4c8f8dec8af029232823
       Docs: man:nginx(8)
    Process: 3188 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 3190 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 3191 (nginx)
      Tasks: 2 (limit: 2301)
     Memory: 2.3M (peak: 2.6M)
        CPU: 37ms
    CGroup: /system.slice/nginx.service
            └─3191 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
               3192 "nginx: worker process"
mohammed@soc-lab:~$ curl -I http://127.0.0.1
HTTP/1.1 200 OK
Server: nginx
Date: Mon, 05 Oct 2026 22:50:17 GMT
Content-Type: text/html
Content-Length: 615
Last-Modified: Mon, 28 Sep 2026 23:33:06 GMT
Connection: keep-alive
ETag: "6abaf932-267"
Accept-Ranges: bytes
mohammed@soc-lab:~$
```